

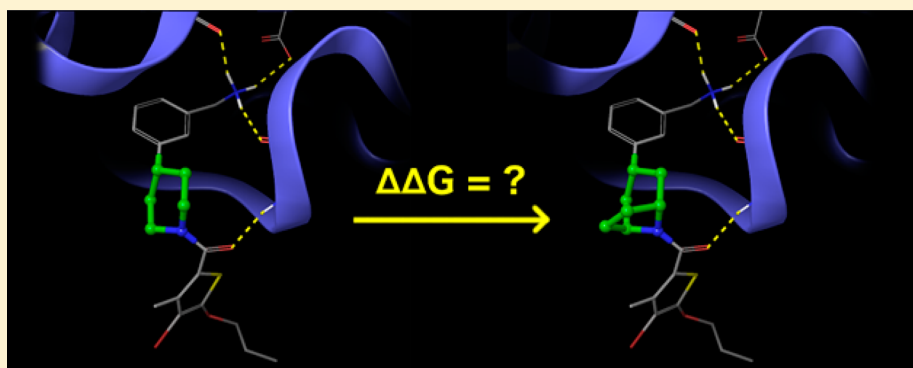
Accurate Modeling of Scaffold Hopping Transformations in Drug Discovery

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S Supporting Information



ABSTRACT: The accurate prediction of protein–ligand binding free energies remains a significant challenge of central importance in computational biophysics and structure-based drug design. Multiple recent advances including the development of greatly improved protein and ligand molecular mechanics force fields, more efficient enhanced sampling methods, and low-cost powerful GPU computing clusters have enabled accurate and reliable predictions of relative protein–ligand binding free energies through the free energy perturbation (FEP) methods. However, the existing FEP methods can only be used to calculate the relative binding free energies for R-group modifications or single-atom modifications and cannot be used to efficiently evaluate scaffold hopping modifications to a lead molecule. Scaffold hopping or core hopping, a very common design strategy in drug discovery projects, is critical not only in the early stages of a discovery campaign where novel active matter must be identified but also in lead optimization where the resolution of a variety of ADME/Tox problems may require identification of a novel core structure. In this paper, we introduce a method that enables theoretically rigorous, yet computationally tractable, relative protein–ligand binding free energy calculations to be pursued for scaffold hopping modifications. We apply the method to six pharmaceutically interesting cases where diverse types of scaffold hopping modifications were required to identify the drug molecules ultimately sent into the clinic. For these six diverse cases, the predicted binding affinities were in close agreement with experiment, demonstrating the wide applicability and the significant impact Core Hopping FEP may provide in drug discovery projects.

INTRODUCTION

A vast number of biological processes depends extensively on protein–ligand binding including signal transduction, cellular metabolism, and gene expression. Similarly, a majority of small molecule drugs also achieve their therapeutic effects through binding to a protein, thereby inhibiting, up-regulating, or down-regulating its function. Therefore, accurate calculation of the binding free energy of a ligand to its target protein remains a grand challenge of central importance in computational biophysics and computer aided drug design.^{1–3} Among the methods to compute protein–ligand binding free energies, free energy perturbation (FEP) calculations performed by way of explicitly solvated molecular dynamics simulations are among the most rigorous and should provide a thermodynamically complete description of the binding event.^{4–7} Multiple recent

advances including the development of greatly improved protein and ligand molecular mechanics force fields,^{8–11} more efficient enhanced sampling methods,^{12–16} and low-cost powerful GPU computing clusters have drastically increased the accuracy, throughput, robustness, and utility of these methods.^{5,17–23} Encouragingly, these methods have begun to generate very promising results, including accurately predicting the binding affinities of ligands in prospective studies and positively impacting industrial drug discovery projects.¹⁷ However, modern relative binding affinity FEP methodologies have a significant limitation. These methods can only be used to calculate the relative binding free energies between molecules with R-group

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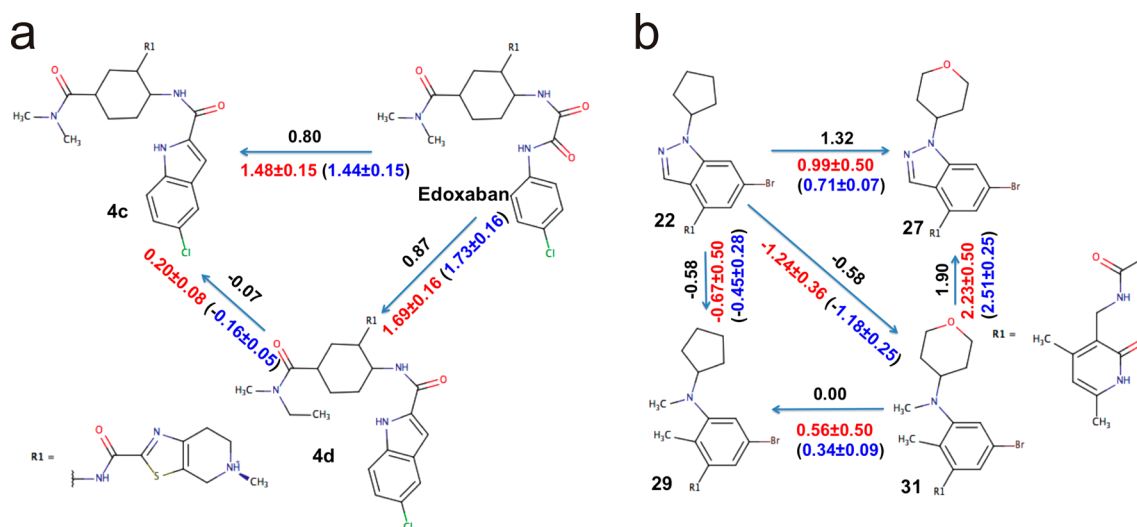


Figure 1. Core Hopping FEP predicted and experimental binding free energies for the Fxa and EZH2 ligands demonstrating the ring opening/closing modifications. Closed thermodynamic cycles are formed in the perturbation pathways. The black numbers are the experimentally observed binding free energy differences. The blue numbers are the relative free energy differences from the direct perturbations and their associated errors using the Bennett Acceptance Ratio (BAR) estimate, and the red numbers are the predicted free energies and their associated error estimates from cycle closure analysis. FEP predicted binding free energies are consistent along different pathways and agree well with experimental data. The nonadditivity effect of EZH2 ligands is also correctly predicted by Core Hopping FEP.

modifications and/or single-atom modifications where the relevant two molecules have the same bond topology.^{24,25}

However, very commonly in drug discovery campaigns, the project teams will pursue scaffold hopping modifications, where the peripheral groups of a molecule are preserved as being crucial for binding to the protein receptor and the changes to the structure are focused on the core region, most often resulting in molecules with different bond topologies.^{26,27} Scaffold hopping is critical not only in the early stages of a discovery campaign where novel active matter must be identified but also in late-stage lead optimization where the resolution of a variety of ADME/Tox problems may require identification of a novel core structure.^{26,27} In addition to scaffold hopping being one of the most valuable medicinal chemistry design strategies, it is also among the riskiest of design strategies since the vast majority of possible scaffold hopping modifications, many of which may require substantial synthetic effort to achieve, will result in a considerable loss of binding potency.^{26,27} Thus, if a scaffold hopping exercise is to succeed, typically a great deal of time-consuming and costly chemistry effort must be committed to identify the few satisfactory scaffolds. However, if one could predict in advance of chemistry resources what new scaffolds were most likely to maintain or improve compound potency, then this chemistry effort could be much more focused leading to much faster identification of satisfactory scaffolds at significantly reduced costs.

We note here that scaffold hopping would typically be considered to include single-atom changes in the core structure of the molecule without changing the bond topology and that the existing FEP method can readily be used to describe such transformations. However, many if not most scaffold hopping modifications do require changing the bond topologies of the core structures of the molecule,^{25–27} in which case existing FEP methods cannot easily be used to evaluate the binding affinity change due to such modifications. Examples of such scaffold hopping modifications include transforming a linear molecule into a ring structure, changing the size of a ring in the core of the

molecule, extending from a single ring into a fused ring or a bridged ring, and so forth.

Unlike R-group modification and/or single-atom transformations, where the core part of the two molecules involved has the same bond topology, the bond topologies of two molecules related by a scaffold hopping transformation are typically different, and the calculation of their relative binding free energies requires the calculation of the free energy associated with the introduction or annihilation of a covalent bond.²⁵ In commonly used classical mechanics force fields, bond stretch interactions between two atoms are represented by a harmonic potential which asymptotically approaches infinity when the distance between the two atoms is very large. This feature of the bond stretch terms implies that a formal singularity exists on the potential energy surface, which may in turn result in numerical instability problems in alchemical FEP and MD simulations, if a conventional FEP approach is taken to describe this term.²⁵

In this article, we present a theoretically rigorous method to calculate the relative protein–ligand binding free energies of pairs of ligands that differ by scaffold hopping modifications. In this new method, which we refer to as Core Hopping FEP, a soft bond stretch potential is used that resolves the singularity and numerical instability problems that hinder alchemical FEP calculations for scaffold hopping modifications.²⁸ We further validate the method on six pharmaceutically interesting drug targets where diverse scaffold hopping strategies were pursued to identify the drug molecules sent to the clinic. In all six cases, the Core Hopping FEP predicted binding free energies agree very well with experimental results. To our knowledge, this is the first attempt to adapt rigorous relative binding free energy calculations to describe the scaffold hopping modifications most relevant to drug discovery.

RESULTS AND DISCUSSIONS

In this section, we will present six pharmaceutically interesting drug targets for which diverse scaffold hopping modifications were employed to optimize the lead compounds, some of which led to the identification of drug candidate molecules sent into the

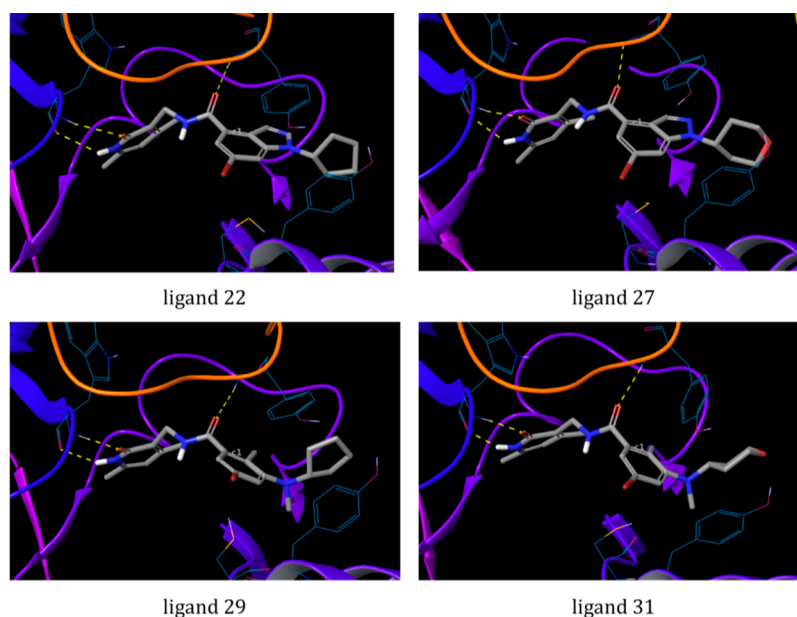


Figure 2. Dominant binding poses of the 4 EZH2 ligands sampled in Core Hopping FEP simulations. In ligands 29 and 31, with the opening of the five-membered ring on bicyclic indazole core, the methyl group on the nitrogen linker are forced to tilt away from the phenol ring because of static clash with the methyl group on the phenol ring. Therefore, the cyclopentane and tetrahydropyran groups in ligands 29 and 31 are pointing in different regions of the binding pocket than the corresponding groups in ligands 22 and 27, resulting in the apparent puzzling structure activity relationship (SAR) and nonadditivity effect.

clinic. We then present the relative protein–ligand binding free energy calculation results on these systems using the Core Hopping FEP method described in the [Methods](#) section and compare the predictions with experimentally observed binding free energy differences. Structures of individual ligands and the protein targets used as the input of the Core Hopping FEP calculations and the output files of all the calculations are included in the [Supporting Information](#). These modifications cover three of the most important and common scaffold hopping transformations pursued in drug discovery projects. These include the transformation from a linear structure to a ring structure in the core of the molecule (ring opening/closing), extending a single ring into a fused ring or bridged ring (ring extension) and changing the size of a ring in the core of the molecule (ring size change). The experimental and Core Hopping FEP predicted relative binding free energies for these scaffold hopping mutations are shown in [Figures 1, 3, 4, and 5](#). In these figures, the black numbers are the experimentally observed binding free energy differences. The blue numbers are the relative free energy differences from the direct perturbations and their associated errors using the Bennett Acceptance Ratio (BAR) estimate, and the red numbers are the predicted free energies and their associated error estimates from cycle closure analysis.

Ring Opening/Closing Transformation. Ring opening/closing transformation here refers to a common type of scaffold hopping modifications whereby a linear structure in the core of the molecule is transformed into a ring structure with the introduction of additional bond(s) (ring closing) or vice versa (ring opening). In a ring closing transformation, at least one additional bond has to be created between two atoms in the targeted linear structure. Similarly, in a ring opening transformation, at least one bond in the ring to be opened has to be annihilated to form the linear structure. Two sets of ligands involving ring opening/closing transformations binding to Factor Xa (FXa)^{29,30} and Enhancer of Zeste Homologue 2

(EZH2)^{31,32} respectively are shown in [Figure 1](#). FXa, a key serine protease that plays an important role in many stages of the coagulation pathway, is an important drug target for the prophylaxis and treatment of thromboembolic diseases. Edoxaban is an oral anticoagulant drug acting as a FXa inhibitor developed by Daiichi Sankyo and approved in 2011 in Japan.³⁰ During the development of Edoxaban, two key backup candidate molecules 4c and 4d were prepared by replacing the carbonyl group with a bicyclic chloroindole, and transformations from edoxaban to 4c and 4d involve ring closing modifications.²⁹ EZH2 catalyzes the addition of a methyl group to histone 3 at lysine 27, and abnormal methylation of this site is found in many cancers.³¹ Tazemetostat is an EZH2 inhibitor developed by Epizyme currently in clinical trials.³¹ Ligands 22, 27, 29, and 31 shown in [Figure 1](#) are four close analogs of Tazemetostat.³² These molecules differ from Tazemetostat by a ring closing modification, and their optimization led to the discovery of Tazemetostat.

The experimental and the Core Hopping FEP predicted relative binding free energies between each pair of ligands in the data set are shown in [Figure 1](#). In these calculations, closed thermodynamic cycles are formed, and cycle closure analysis methods detailed in prior publications^{17,18} are used to obtain the statistically optimal estimates of the free energies and the sampling errors of the calculations. The black numbers are the experimentally observed binding free energy differences. The blue numbers are the relative free energy differences from the direct perturbations and their associated errors using the Bennett Acceptance Ratio (BAR) estimate, and the red numbers are the predicted free energies and their associated error estimates from cycle closure analysis. The BAR error only captures the statistical error of the free energy estimate from the direct perturbation based on the assumption that all the important regions in phase space have been sampled in that perturbation. The cycle closure error takes into consideration the multiple pathways connecting the same pair of ligands and usually provides a more reliable

estimate about the sampling error in the calculation. In both cases, the hysteresis values of the closed thermodynamic cycles are very small, indicating that the free energy calculations are likely converged.

For FXa ligands, with the annihilation of carbonyl group in Edoxaban and the formation of the five-membered-fused ring in 4c, the experimental binding potency decreases slightly, with a K_i of 0.6 nM for Edoxaban and a IC_{50} of 2.3 nM for 4c, approximately a 0.8 kcal/mol decrease in binding potency.³⁰ The Core Hopping FEP correctly predicts that ligands 4c binds more weakly than Edoxaban, with a predicted $\Delta\Delta G$ of 1.48 kcal/mol. The addition of an extra methyl group at the amide tail in ligand 4d does not materially affect the free energy,²⁹ which is also correctly predicted by Core Hopping FEP.

For EZH2 ligands, opening the five-membered ring on the bicyclic indazole core and transforming the cyclopentane to tetrahydropyran attached to an indazole demonstrates a strong nonadditivity effect. Specifically, when opening the indazole core with a cyclopentane attached (from ligand 22 to 29) only a modest increase in the binding potency (0.58 kcal/mol) is observed; however, the same transformation when the tetrahydropyran is attached (from ligand 27 to 31) dramatically increases the binding potency by about 1.9 kcal/mol.³² This strong nonadditivity is correctly predicted by the Core Hopping FEP calculations. FEP predicts a ~ 2.23 kcal/mol potency boost from ligand 27 to 31 and only a modest potency improvement of ~ 0.67 kcal/mol from ligand 22 to 29.

The dominant binding modes sampled in Core Hopping FEP simulations shown in Figure 2 provide some insight regarding the source of the nonadditivity. In ligands 29 and 31, with the opening of the five-membered ring on the bicyclic indazole core, the methyl group on the nitrogen linker is forced to tilt away from the phenol ring because of steric clash with the methyl group on the phenol ring. Therefore, the cyclopentane and tetrahydropyran groups in ligands 29 and 31 are pointing in different regions of the binding pocket than the corresponding groups in ligands 22 and 27, explaining the initially puzzling structure activity relationship (SAR).

Ring Extension Transformation. Ring extension transformation refers to another common type of scaffold hopping modification in which additional atoms are introduced in the core of the molecule, forming one or more additional rings fused with existing ring(s). For this type of transformation, a bond in the additional ring of the second molecule has to be annihilated to avoid the coupling between the additional ring and the existing ring of the first molecule. Two sets of ligands involving ring extension transformations binding to Beta-tryptase (TPSB2)³³ and Beta-secretase 1 (Bace1)³⁴ respectively are shown in Figure 3. TPSB2 is a trypsin-like protease reported to have a strong correlation with inflammatory and allergic disorders. Compounds 1 and 2 are both TPSB2 inhibitors, and compound 2 is reported to have less problematic off-target effects than compound 1 because the tropanylamide core is more rigid than the piperidinylamide core.³³ Transformation from compound 1 to compound 2 involves transforming a six-membered ring into a bridged ring, a typical scaffold hopping modification in a drug discovery campaign. Bace1 has been aggressively pursued as a key target for the treatment of Alzheimer's disease. Extensive recent work carried out by Lilly has resulted in the discovery of a series of nanomolar Bace1 inhibitors with a novel bicyclic thiazine ring, of which compounds 6, 7, and 31 shown in Figure 3 are representative.³⁴ Transformations from compounds 6 and 31

to compound 7 both involve transforming a bicyclic thiazine ring into a 6–6–6 bridged ring.

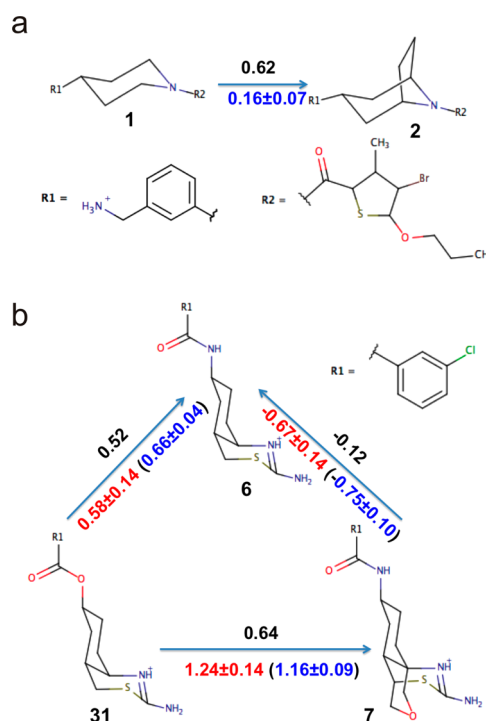


Figure 3. Core Hopping FEP predicted and experimental binding free energies for the TPSB2 and Bace1 ligands demonstrating the ring extension modifications. Closed thermodynamic cycles are formed in the perturbation pathways. The black numbers are the experimentally observed binding free energy differences. The blue numbers are the relative free energy differences from the direct perturbations and their associated errors using the Bennett Acceptance Ratio (BAR) estimate, and the red numbers are the predicted free energies and their associated error estimates from cycle closure analysis. FEP predicted binding free energies are consistent along different pathways and agree well with experimental data.

The experimental and the Core Hopping FEP predicted relative binding free energies for these ligands are shown in Figure 3. For TPSB2 ligands, transformation from the piperidinylamide core to the tropanylamide core slightly decreases the binding potency by ~ 0.62 kcal/mol,³³ which is correctly predicted by Core Hopping FEP with a predicted value of 0.16 kcal/mol. For Bace1 ligands, transformation from the bicyclic ring of ligand 6 into bridged ring of ligand 7 does not affect the binding potency significantly,³⁴ and the Core Hopping FEP calculations also predict ligands 6 and 7 to be roughly equally potent. The slight increase of the binding potency from the amide linker in ligand 6 to the ester linker in ligand 31 is also correctly predicted.

Ring Size Change Transformation. Ring size change transformation is another common type of scaffold hopping modification where one or more rings in the core of the initial and final molecules have different ring sizes. For this type of transformations, one bond in the first molecule needs to be annihilated, while a different bond in the second molecule is simultaneously formed. Four ligands involving ring size change transformations binding to Estrogen Receptor alpha (ERa)³⁵ are shown in Figure 4. Estrogen receptors (ER) are members of the steroid nuclear hormone receptor family, and selective ER modulators (selectivity between ERa and ERb) might have

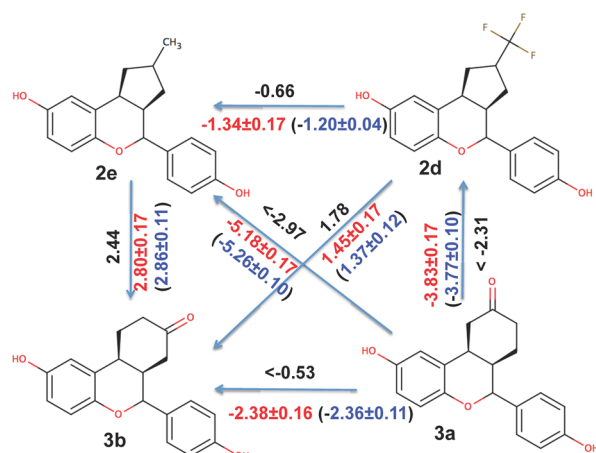


Figure 4. Core Hopping FEP predicted and experimental binding free energies for the ERA ligands demonstrating the ring size change modifications. Closed thermodynamic cycles are formed in the perturbation pathways. The black numbers are the experimentally observed binding free energy differences. The blue numbers are the relative free energy differences from the direct perturbations and their associated errors using the Bennett Acceptance Ratio (BAR) estimate, and the red numbers are the predicted free energies and their associated error estimates from cycle closure analysis. FEP predicted binding free energies are consistent along different pathways and agree well with experimental data.

potential therapeutic benefits in multiple diseases.³⁵ Recent work by Lilly has resulted in ERb selective ligands involving ring size changes in the core part of the ligands as shown in Figure 4.³⁵

The experimental and Core Hopping FEP predicted relative binding free energies among these ligands are shown in Figure 4. A few closed thermodynamic cycles are formed in the perturbation pathways, and the hysteresis values in all the cycles are small, indicating that the free energy results are likely converged. Experimentally, expanding the ring size from 5 to 6 in going from ligand 2e to ligand 3b dramatically decreases the

binding potency by ~ 2.44 kcal/mol,³⁵ and Core Hopping FEP correctly predicted this significant drop in potency with a predicted value of 2.80 kcal/mol. Switching the position of the carbonyl group on the six-membered ring from ligand 3b to 3a further decreases the binding potency, resulting in an undetectable IC₅₀ of greater than 1000 nM for ligand 3a.³⁵ This effect is also correctly predicted by Core Hopping FEP, with a predicted potency drop of more than 5 kcal/mol in going from ligand 2e to ligand 3a.

Mixture of All Types of Core Hopping and R-Group Modifications. The 5 ligands binding to checkpoint kinase 1 (CHK1)³⁶ shown in Figure 5 demonstrate a mixture of all types of core hopping and R-group modifications that a typical lead optimization program might explore. CHK1 is a very important drug target for anticancer therapies, and extensive work on CHK1 inhibition has been undertaken by a number of groups. The 5 CHK1 ligands shown in Figure 5 are taken from a recent drug discovery program.³⁶ Modifications that can be observed³⁶ include ring opening/closing transformation (ligand 19 to 21), ring extension (ligand 17 to 20, 17 to 21), ring size change (ligand 20 to 21), heteroatom change (ligand 1 to 20), and R-group modification (ligand 17 to 19).

The FEP predicted relative binding free energies, along with the experimental data, are shown in Figure 5. The FEP predicted binding free energies through different pathways are highly consistent with each other and agree well with experimental data. These results suggest that, with the Core Hopping FEP methodology we have introduced here, accurate predictions of the relative binding free energies can be obtained for the vast majority of types of chemical modifications a drug discovery program might explore.

The correlation plot between the Core Hopping FEP predicted relative binding free energies, and their corresponding experimental results for the 18 perturbations involving scaffold hopping modifications for all six systems studied here is shown in Figure 6. The two perturbations involving ligand 3a for Era (3a $\rightarrow$ 2d and 3a $\rightarrow$ 2e) were not included in this plot because

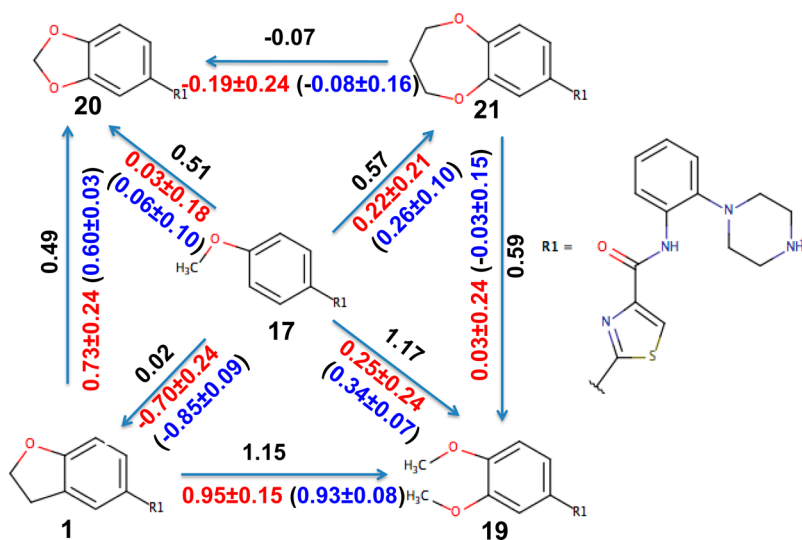


Figure 5. Core Hopping FEP predicted and experimental binding free energies for the 5 Chk1 ligands involving a mixture of all types of perturbations, including the regular R-group modification, ring size change perturbation, ring open/closing perturbation, and ring extension perturbation. Closed thermodynamic cycles are formed in the perturbation pathways. The black numbers are the experimentally observed binding free energy differences. The blue numbers are the relative free energy differences from the direct perturbations and their associated errors using the Bennett Acceptance Ratio (BAR) estimate, and the red numbers are the predicted free energies and their associated error estimates from cycle closure analysis. FEP predicted binding free energies are consistent along different pathways and agree well with experimental data.

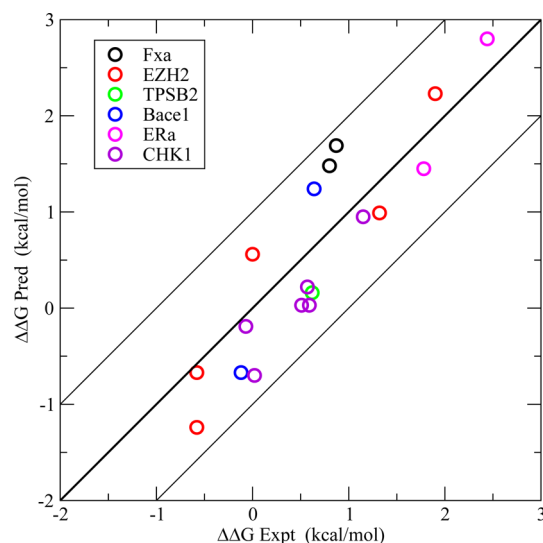


Figure 6. Comparison between the Core Hopping FEP predicted relative binding free energies and their corresponding experimental data for all the scaffold hopping mutations in the six systems. The Core Hopping FEP predicted relative binding free energies agree very well with the experimental data, with a root-mean-square error (RMSE) of 0.50 kcal/mol and a R^2 value of 0.81.

the exact experimental binding free energy for 3a is unknown. For this limited number of validation testing, the Core Hopping FEP predicted relative binding free energies agree very well with the experimental results, with a root-mean-square error (RMSE) of 0.50 kcal/mol and a R^2 value of 0.81. All the perturbations tested here only have one bond stretch term that has to be annihilated where the general scaffold hopping modifications might involve much larger perturbations. Therefore, a RMSE of 0.50 kcal/mol seen here might be a lower bound of the real error for large-scale applications of Core Hopping FEP in scaffold modifications in real drug discovery projects.

Comparison between Regular R-Group Modification FEP and Core Hopping FEP. It is interesting to note here that if the ring opening/closing, ring extension, and ring size change transformations are occurring in the peripheral part of the ligands, like the CHK1 ligands shown in Figure 5, rather than directly changing the ring topology through Core Hopping FEP, the regular R-group modification FEP can also be employed to estimate the relative binding free energies for such ligands. For example, for perturbation from ligand 17 to ligand 20 binding to CHK1, the relative binding free energy can be estimated in an R-group modification FEP by annihilating the methoxybenzene ring in ligand 17 at the same time when the bicyclic ring in ligand 20 is grown. Similar strategies can be employed for the other perturbations in the CHK1 system. The R-group modification FEP predicted relative binding free energies for CHK1 ligands are shown in Figure S1 in the Supporting Information. The Core Hopping FEP predicted relative binding free energies also agree very well with the extensively validated R-group modification FEP results, with a slight improvement in the cycle closure convergence error (average cycle closure converge error of 0.21 for Core Hopping FEP versus 0.26 for R-group modification FEP). These results indicate that the Core Hopping FEP method introduced here achieves the same level of accuracy as the more extensively validated R-group modification FEP¹⁷ with a more efficient perturbation scheme leading to slightly more converged predictions.

However, if the ring topology change is occurring in the middle part of the ligands, like the TP5B2 ligands, or if the core of the two molecules using the R group modification FEP is very small, like the Bace1 ligands, the regular R group modification FEP is not able to obtain reliable predictions for such perturbations. The R group modification FEP predicted relative binding free energies for all the systems are included in Supporting Information. For the TP5B2 and the Bace1 systems, the ligands unbind in the intermediate lambda windows using the regular R group modification FEP, resulting in a much larger sampling error and much worse free energy predictions as compared to Core Hopping FEP results. The sampling errors using the R group modification FEP for other systems are also larger than the corresponding Core Hopping FEP results, resulting in an overall RMSE of 0.83 kcal/mol for R group modification FEP versus 0.50 kcal/mol for Core Hopping FEP.

CONCLUSIONS

One of the most important problems in computational drug design is how to accurately and reliably calculate protein–ligand binding free energies. Due to the continuous and ongoing improvements in protein and ligand molecular mechanics force fields, more efficient enhanced sampling methods, and low-cost powerful GPU computing clusters, accurate and reliable predictions of protein–ligand binding free energies via the use of theoretically rigorous FEP methods have begun to play an ever expanding role in drug discovery campaigns. While the existing FEP methodology can be used to calculate the relative binding free energies for molecules involving R-group modifications or single-atom modification, it historically has not been applicable to scaffold hopping modifications where the core structure of the relevant molecules has different bond topologies.

The extension of relative binding affinity FEP calculations to this important problem area requires the calculation of the free energy to introduce or annihilate a bond. In this article, we have presented a soft bond stretch potential that makes FEP calculation for this type of chemical modifications more feasible. We further show that the soft bond stretch potential resolves the singularity and numerical instability problems associated with the conventional alchemical free energy calculation methods for bond introduction and annihilation.

To demonstrate the validity of the approach, we have applied the presented Core Hopping FEP method to six pharmaceutically interesting drug targets involving chemically diverse scaffold hopping modifications in the reported ligand series. Some of the ligands are commercially available drugs or currently in clinical trials. The scaffold hopping modification studied here covers three of the most important core hopping modifications a typical drug discovery program might explore, including ring opening/closing, ring-extension, and ring-size change. In all six cases, the Core Hopping FEP method predicted relative binding free energies are consistent with the experimental data, with a RMSE of 0.50 kcal/mol and a R^2 value of 0.81.

Scaffold hopping modifications will most often result in change in the topology of the pharmacophore, and thus a considerable loss of activity should be expected for most of the new scaffolds.²⁶ Accurate prediction of the potency changes due to scaffold hopping modifications will therefore eliminate the scaffolds that are not likely to meet the targeted potency and enable chemists to pursue synthetic directions that would otherwise be considered as too risky. Given the great importance scaffold hopping modifications hold in drug discovery projects, we anticipate the ability to model the outcomes of such

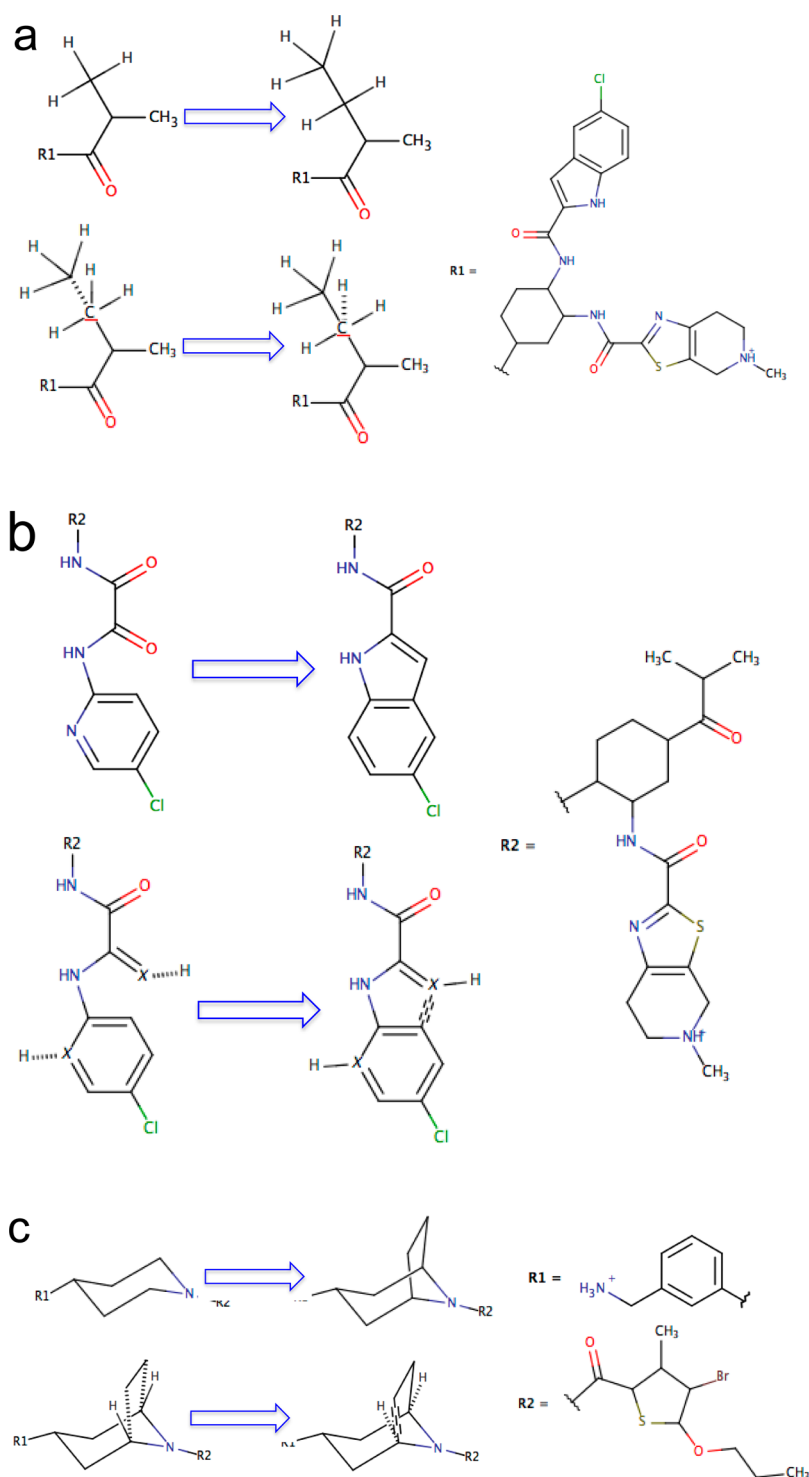


Figure 7. Example modifications demonstrating the fundamental differences between regular R-group modification FEP and core-hopping FEP. The structure of the mixed molecule used in FEP simulation for each case is shown at the bottom row. The dashed lines represent the bonded attachments between the core part of the molecule and the dummy atom groups. Part a: an example regular R-group modification transformation for a pair of Fxa ligands where a hydrogen atom is mutated to a methyl group. In the initial state, the extra methyl group in the second molecule becomes dummy atoms, and, in the final state, the extra hydrogen atom in the first molecule becomes a dummy atom. In regular R-group modification FEP, each dummy atom group only has one bonded attachment with the core part of the molecule, and the bonded connections between the physical atoms in the ligands remain unchanged. Part b: an example core-hopping modification for a pair of Fxa ligands where an extra ring is formed in the core part of the molecule. The double dashed line represents the extra bond between two physical atoms in the ligand in the second molecule. Part c: an example core-hopping modification for a pair of tpsb2 ligands where a six-membered ring is transformed into a bridged ring. The dummy atom group has two bonded attachments with the core part of the molecule, and the free energy for the formation of one of these two bonds (the double dashed line in the second molecule) has to be calculated. In core-hopping modifications, either extra bonds between physical atoms in the ligands are formed or annihilated or/and the dummy atom groups have two bonded attachment with the core part of the molecule, and explicit calculation of the free energy to form or break a bond is required.

modifications will provide tremendous value to the discovery teams attempting to design novel small molecule therapies.

We further anticipate the Core Hopping FEP methodology introduced here may have significant implications for multiple other research areas not detailed in this work. For example, many human diseases are caused by abnormal protein mutations, and calculation of the effect of a mutation on the biological function of a protein may require relative free energy calculation between the wild type and mutant form. The Core Hopping FEP methodology introduced in this paper may enable modeling of proline residue mutations or residue insertion or deletion mutations, which have traditionally not been possible due to the associated bond topology modification related challenges. Likewise, cyclic peptides have emerged as an important class of natural products and have important applications in medicine and biology. Investigations of why structurally similar cyclic peptides have different biological or therapeutic effects may require the Core Hopping FEP methodology should the cyclic peptides differ in their sequence length. Given these possibilities, we expect the Core Hopping FEP method introduced here to have a variety of future applications in multiple research areas.

METHODS

We begin with a brief discussion of the underlying theory of alchemical free energy calculations as it is commonly adapted to describe relative protein–ligand binding free energies and illustrate the fundamental differences between R-group modification FEP and Core Hopping FEP, which motivated the formulation of the soft bond stretch potential. In alchemical free energy calculations, the relative free energy difference between two states, with Hamiltonians H_0 and H_1 respectively, is calculated by smoothly changing the Hamiltonian from H_0 to H_1 , usually through a linear combination of the two Hamiltonians that is controlled by a thermodynamic coupling parameter λ , where $H(\lambda = 0) = H_0$ and $H(\lambda = 1) = H_1$.^{4,6} The free energy difference between the two states is determined using thermodynamic integration (TI), free energy perturbation (FEP), or related methods. Further, in relative protein–ligand binding free energy calculations, two alchemical free energy simulations are performed. The first simulation transforms the first ligand into the second ligand in the protein binding pocket, and the second simulation transforms the ligand in the bulk solvent. The difference between the free energies of these two transformation processes is formally the relative binding free energy of the two ligands.^{1,4}

In the general case, as in the three examples shown in Figure 7, the two ligands may have a different number of atoms and different bond topologies. Therefore, dummy atoms are introduced in relative protein–ligand binding free energy calculations to enable the calculation of the energy difference between a pair of thermodynamic states with different coupling parameters or the derivative of the Hamiltonian with respect to the thermodynamic coupling parameter, as required for the calculation of relative free energy via FEP or TI.²⁴ Depending on how the dummy atoms are introduced in FEP simulations, single topology and dual topology methods may be employed.²⁴ While the single topology FEP method shares as many atoms as possible between the two molecules and minimizes the number of dummy atoms introduced, the dual topology FEP method introduces additional dummy atoms as needed. These dummy atoms do not have interactions with the protein or with waters, but they must retain some bond stretch interactions with the rest of the molecule to minimize the configurational space sample

when the dummy atoms are decoupled from the physical system. A critical requirement in the formulation of any valid relative protein–ligand binding free energy calculation method is that the dummy atom motional contributions to the free energy difference in the two legs of alchemical free energy calculations must exactly cancel. Otherwise, unbalanced contributions from the dummy atoms to the free energy may introduce significant error in the calculated relative binding free energy.^{24,25}

The fundamental difference between regular R-group modification FEP and Core-hopping FEP is that R-group modification FEP does not require the calculation of the free energy to form or to break a bond which is necessary for Core Hopping FEP.²⁵ This can be best demonstrated in the example shown in Figure 7. In R-group modification FEP, as in the example shown in Figure 7a for a pair of FXa ligands involving a R-group modification, the bond topologies of the core of the two molecules are the same and each functional group of dummy atoms has only a single point of attachment with the rest of the molecule. In this case, it is trivial to prove that the bond stretch interaction between the anchoring atom of dummy atom group and the anchoring atom of the core part of the molecule does not need to be in the perturbed part of the Hamiltonian.^{24,25} However, in Core Hopping FEP as the example shown in Figure 7b for a pair of FXa ligands involving a ring open/closing transformation and the example shown in Figure 7c for a pair of TP5B2 ligands involving a ring extension transformation, either extra bond(s) is/are formed in the core of the molecule (Figure 7b), or/and one of the dummy atom groups has two points of attachment with the rest of the molecule (Figure 7c). In these types of scaffold hopping transformations, we can rigorously prove that inclusion of all bonded stretch interactions present in either ligands in the unperturbed part of the Hamiltonian will introduce artificial coupling to the physical atoms in the ligand and introduce errors in the free energy results.²⁵ Further, previous work has demonstrated that improper treatment of the bonded interactions between the dummy atoms and the physical atoms in the ligand can introduce a large error in the free energy results.²⁵ Therefore, Core Hopping FEP requires the explicit calculation of the free energy to introduce or annihilate (a) bonds in the two legs of alchemical free energy calculations. This is a difficult problem, which no one to our knowledge has effectively addressed. In what follows, we will explain why it is difficult to calculate the free energy to form or break a bond using conventional FEP, and how the soft bond stretch potential addresses this problem.

In the commonly used molecular mechanics force fields,^{8–11} the bond stretch interaction between the two atoms is represented by a harmonic potential of the form

$$U_{bs}(r) = \frac{1}{2}k(r - r_0)^2$$

where k is the force constant, and r_0 is the equilibrium distance.

In conventional alchemical free energy methods, the following bonded stretch term would be used in the coupling Hamiltonian where the harmonic potential is linearly scaled by a thermodynamic coupling parameter λ :

$$U_{bs}(\lambda, r) = \frac{1}{2}\lambda k(r - r_0)^2$$

Using thermodynamic integration, it is easy to show that

$$F(\lambda = 1) - F(\lambda = 0) = \int_0^1 d\lambda \frac{\partial F(\lambda)}{\partial \lambda} = \int_0^1 d\lambda \left\langle \frac{1}{2}k(r - r_0)^2 \right\rangle_\lambda$$

The ensemble average in the integrand of the above equation approaches infinity when r is very large. In the limit when λ approaches zero, and where there is no bond stretch interaction between the two atoms, the two atoms will tend to be far away from each other and the above ensemble average will approach infinity. This asymptotic behavior will naturally lead to numerical instability problems in the calculation of the above integral. In molecular dynamics simulations, the distance between the two atoms is of course limited by the size of the simulation box, and Free Energy Perturbation⁶ or Bennett Acceptance Ratio³⁷ formula instead of thermodynamic integration formula can be used to calculate the free energy difference, which may alleviate some of these difficulties. However, since the integrand in the above equation is unbounded, these numerical instabilities cannot be entirely eliminated.

Furthermore, using the conventional linear coupling of the bond stretch interaction, the potential energy is undefined when λ approaches 0 and r approaches infinity. The limiting value depends on how fast λ approaches 0 and how fast r approaches infinity, i.e.:

$$\lim_{r \rightarrow \infty} \lim_{\lambda \rightarrow 0} U_{bs}(\lambda, r) = \lim_{r \rightarrow \infty} \lim_{\lambda \rightarrow 0} \frac{1}{2} \lambda k (r - r_0)^2 \quad \text{Diverges}$$

To obtain a thermodynamic pathway connecting the two physical end-states with and without the bond stretch interaction that allows stable and efficient simulations from which reliable free energies involving the annihilation and introduction of a harmonic bond between two atoms can be determined, we suggest the following coupling potential where the bonded stretch interactions between the two atoms are fully turned on and off when the coupling parameter λ changes between 1 and 0:

$$U_{bs}(\lambda, r) = \frac{1}{2} \lambda k (r - r_0)^2 \frac{1}{1 + \alpha(1 - \lambda)(r - r_0)^2}$$

We call the above potential the soft bond stretch potential, and α , the soft bond parameter, is a positive number that can be adjusted to maximize the phase space overlap between neighboring lambda windows. It is trivial to show that the soft bond stretch interaction correctly recovers the two physical end states when $\lambda = 0$ and $\lambda = 1$, i.e.

$$U_{bs}(r, \lambda = 1) = \frac{1}{2} k (r - r_0)^2$$

$$U_{bs}(r, \lambda = 0) = 0$$

$$U_{bs}(r \rightarrow \infty, \lambda) = \frac{k\lambda}{2\alpha(1 - \lambda)} \text{ when } \lambda \neq 1$$

The introduction of $\alpha(1 - \lambda)(r - r_0)^2$ in the denominator changes the harmonic interaction into a soft bond interaction when λ is less than 1 (the potential energy is bounded when r approaches infinity), and the function recovers the exact harmonic potential when $\lambda = 1$.

In [Appendix A](#), we will discuss the mathematical properties of the soft bond stretch potential functional form and explain how the singularity and numerical instability problems are resolved by the choice of this functional form. In [Appendix B](#), we prove that the free energy for the thermodynamic state at $\lambda = 0$ is independent of the particular choice of the value of soft bond parameter α .

Another technical difficulty regarding Core Hopping FEP is related to the change of the exclusion list between the pair of molecules. Note, an exclusion list, in nearly all modern MD

implementations, is used to control the pairs of atoms for which van der Waals and electrostatic and interactions will be computed. In R-group modification FEP, the bond topology of the core of the molecules remains unchanged ([Figure 7a](#)), and the exclusion list does not change between the two physical end-states. For Core Hopping FEP, as in the example shown in [Figure 7b](#), the bond topology among the physical atoms changes between the two end-states, resulting in the change of exclusion list as well. In the commonly used molecular mechanics force fields, the interactions between atoms connected by less than 3 bonds away are fully described by the bonded interactions, and they do not have nonbonded electrostatic and van der Waals interactions.^{8–11} These atom pairs are included in an “exclusion list” to avoid the calculation of such terms. In the example shown in [Figure 7b](#), while the nonbonded interactions between the two atoms connected by the extra bond in the second molecule are excluded for the second molecule with the formation of a five-membered ring, they are not in the exclusion list for the first molecule and should have regular nonbonded interactions. Special care must be taken in the implementation of Core Hopping FEP to reflect the change of the exclusion list between the two end-states. In the implementation presented here, a static exclusion list, which is a union of the exclusion list for both end-states, is used across all the lambda windows. For those pairs of atoms that should not be excluded in either end point, extra pair interactions are added to compensate the loss of nonbonded interactions, and these extra pair interactions are slowly turned off to zero in the other end-state.

Details of the Simulation Protocol. The FEP+ product in the Schrodinger software suite was used to set up and run all the simulations.³⁸ The starting structures of all the simulations are taken from the PDB structures, with PDB IDs 2ei8, 5ij7, 3v7t, 4zsp, 2q70, and 3u9n, respectively. The protein structures were prepared with the Protein Preparation Wizard³⁹ during which the protonation states were assigned assuming a pH of 7. In all the systems except EZH2, the PDB structure is cocrystallized with one of the ligands studied by Core Hopping FEP, while for EZH2 the crystal structure is cocrystallized with a close analog of ligands studied here. The input structures of the ligands were obtained by superimposing the ligands with the cocrystal ligands in the PDB. The OPLS3 force fields⁸ were used for the proteins and ligands along with the SPC model⁴⁰ for water.⁴⁰ The systems are relaxed and equilibrated with the default relaxation protocol implemented in FEP+ as reported in previous publications^{17,18} before the production lambda hopping FEP simulations.

In the production FEP simulations, a total number of 16 lambda windows were used for the Core Hopping perturbations, and the default 12 lambda windows are used for the R-group perturbations. The simulations lasted 5 ns for each replica window for both the complex and solvent legs. The electrostatic interactions were turned off before the van der Waals interactions and soft core Lennard-Jones (LJ) interactions⁴¹ were used to avoid the singularity and numerical instability problems for turning on/off the van der Waals interaction. The soft bond stretch potential introduced in the above section was used in the Core Hopping perturbations to smoothly turn on and/or off the bond stretch interactions that were introduced or annihilated during the transformations. In principle, any reasonable value of soft bond parameter (any positive constant) could be used to converge the free energy calculations. In practice, the soft bond parameter α should be tuned with the lambda schedules to ensure sufficient phase overlap between neighboring lambda windows. In our case, the soft bond parameter α is set to 2, and

we found this setting to be sufficient for all these calculations and expect it to be generally transformable to other perturbations as well. The thermodynamic coupling parameter λ for the soft bond stretch potential is scaled between 1 and 0 with approximately geometric series spacing except when λ is close to 0. In Appendix C, we show that this set of λ values will generate nearly optimal intermediate states with approximately equal thermodynamic length, which maximizes the efficiency of free energy calculations. The relative free energy difference between neighboring λ windows was calculated using the Bennett Acceptance Ratio (BAR) method,³⁷ and the sum of free energy differences between neighboring λ windows gives the relative free energy difference between the physical end states.

APPENDIX A

How the Soft Bond Stretch Potential Resolves the Singularity and Numerical Instability Problem in Core Hopping FEP

The soft bond stretch potential as introduced in the Methods section of the paper has a few important mathematical properties, which resolves the singularity and numerical instability problem for the calculation of free energy to annihilate or introduce a bond. In this section, we are going to list these important properties. This section is taken from the patent in ref 28.

Property 1. The soft bond stretch potential does not have any singular regions for all values of λ between 0 and 1 and for all values of r . This is obvious from the definition of the soft bond stretch potential.

The soft bond stretch potential for a model system with a force constant of $k = 20 \text{ kcal/mol}\cdot\text{\AA}^2$ and the soft bond parameter $\alpha = 2$ at different values of λ are given in Figure 8 (upper panel). It is clear that the potential is continuous for all values of λ and r . It changes slowly from a harmonic potential at $\lambda = 1$ to a soft bond potential at intermediate values of λ and goes to 0 at $\lambda = 0$.

Property 2. The derivative of the free energy associated with the soft bond potential with respect to the thermodynamic coupling parameter λ is bounded for all values of r .

$$\frac{\partial F_{\text{sb}}(\lambda)}{\partial \lambda} = \left\langle \frac{\partial U_{\text{sb}}(\lambda)}{\partial \lambda} \right\rangle_{\lambda} = \left\langle \frac{1}{2} k (r - r_0)^2 \frac{1 + \alpha(r - r_0)^2}{(1 + (1 - \lambda)\alpha(r - r_0)^2)^2} \right\rangle_{\lambda} \quad (\text{A.1})$$

As discussed in the Methods section, it is $\frac{\partial F}{\partial \lambda}$ which determines the numerical stability and accuracy of the free energy calculations. In the above formulation, when $\lambda \in [0, 1)$, the thermodynamic property to be averaged in the bracket in eq A.1 is continuous and bounded for all values of r . When $\lambda = 1$, the soft bond potential recovers the harmonic potential, and only phase space regions where r is close to r_0 are sampled and taken into the ensemble average. In the phase space regions where r is close to r_0 , the integrand in eq A.1 is also bounded. Therefore, the quantity in the bracket of eq A.1 is bounded for all values of λ between 0 and 1. Since the derivative of the free energy associated with the soft bond potential with respect to the thermodynamic coupling parameter λ does not have any singular region for all values of λ between 0 and 1, accurate and reliable free energy results can be obtained using the above soft bond potential functional form. As an example, the derivative of the soft bond potential with respect to the coupling parameter λ at $\lambda = 0.5$ for a model system with a force constant of $k = 20 \text{ kcal/mol}\cdot\text{\AA}^2$ and the soft bond parameter $\alpha = 2$ is plotted in Figure 8 (lower

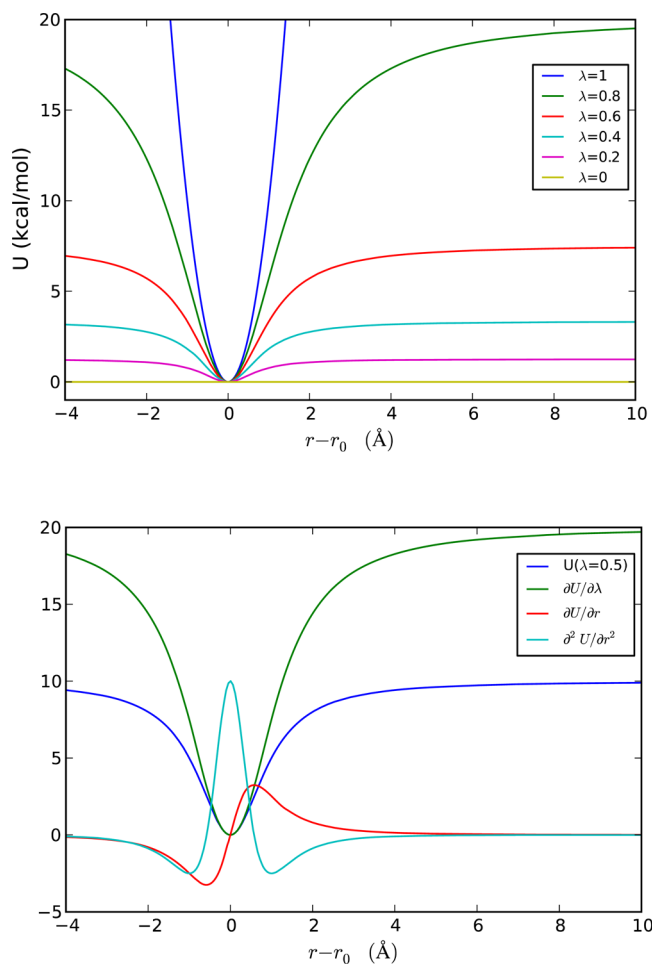


Figure 8. Soft bond stretch potential for a model system with a force constant of $k = 20 \text{ kcal/mol}\cdot\text{\AA}^2$ and the soft bond parameter of 2. Upper: the soft bond stretch potentials at different values of the thermodynamic coupling parameter λ . The soft bond stretch potential smoothly changes from the harmonic potential at $\lambda = 1$ to 0 at $\lambda = 0$, and there is no singular region in the potential at all values of r and all values of λ . Lower: the derivative of the soft bond stretch potential with respect to the thermodynamic coupling parameter λ and the first two derivatives of the soft bond potential with respect to the interparticle distance r . All these derivatives are continuous and bounded allowing stable molecular simulations to be performed to calculate the free energy to introduce or to annihilate a bond.

panel). Clearly, $\frac{\partial F}{\partial \lambda}$ does not have any singular region for all values of λ and r , allowing reliable and accurate free energy calculations.

Property 3. Both the first derivative and the second derivative of the soft bond potential with respect to the interparticle distance r are continuous and bounded, i.e.

$$\begin{aligned} \frac{\partial U_{\text{sb}}(\lambda, r)}{\partial r} &= k\lambda \frac{r - r_0}{(1 + (1 - \lambda)\alpha(r - r_0)^2)^2} \\ \frac{\partial^2 U_{\text{sb}}(\lambda, r)}{\partial r^2} &= k\lambda \frac{1 - 3(1 - \lambda)\alpha(r - r_0)^2}{(1 + (1 - \lambda)\alpha(r - r_0)^2)^3} \end{aligned} \quad (\text{A.2})$$

When $\lambda = 1$, the soft bond stretch potential recovers the harmonic potential, and only phase space regions where r is close to r_0 are sampled. Thus, the first and the second derivatives of the potential with respect to the interparticle distance r are continuous and bounded in the physically accessible phase

space regions when $\lambda = 1$. When $\lambda \in [0, 1)$, the first and the second derivative of the soft bond potential with respect to the interparticle distance r are also continuous and bounded, as can be easily seen from eq A.2. Therefore, the forces and accelerations on the atoms are always continuous and bounded for all values of λ between 0 and 1, allowing stable molecular dynamics simulations to be performed for all values of λ . As an example, the first and second derivatives of the soft bond potential with respect to the interparticle distance r at $\lambda = 0.5$ for a model system with a force constant of $k = 20 \text{ kcal/mol} \cdot \text{\AA}^2$ and the soft bond parameter $\alpha = 2$ are plotted in Figure 8 (lower panel). It can be seen that those derivatives are continuous over the whole space, allowing stable molecular dynamics simulations.

APPENDIX B

The Free Energy of the Thermodynamic State at $\lambda = 0$ Is Independent of the Soft Bond Parameter α

As described in Methods section, in the conventional coupling Hamiltonian, the potential energy is not well-defined when λ approaches 0 and r approaches infinity. In contrast, in the soft bond stretch potential, the potential energy is well-defined for all values of λ between 0 and 1 and all values of r . The conventional coupling Hamiltonian is a special case of the soft bond potential where $\alpha = 0$. While the conventional coupling Hamiltonian and the soft bond potential reach the same end state when $\lambda = 1$, the initial states when $\lambda = 0$ are different depending on the value of the soft bond parameter α . In this section, we will

prove that the free energy of the initial state at $\lambda = 0$ does not depend on the soft bond parameter α . This proof was first given in ref 28 and repeated here.

Consider a Hamiltonian with the following potential energy term:

$$U_{\text{sb}}(\lambda, r; \alpha) = \frac{1}{2} \lambda k (r - r_0)^2 \frac{1}{1 + \alpha(1 - \lambda)(r - r_0)^2}$$

When $\alpha = 0$, it becomes the conventional coupling harmonic potential, and when $\alpha > 0$, it becomes the soft bond potential. We need to prove that

$$F(\alpha = 0, \lambda = 0) = F(\alpha = \alpha' > 0, \lambda = 0)$$

Note that

$$F(\alpha = \alpha' > 0, \lambda = 0) - F(\alpha = 0, \lambda = 0) = \int_0^{\alpha'} d\alpha \lim_{\lambda \rightarrow 0} \frac{\partial F(\alpha, \lambda)}{\partial \alpha}$$

where

$$\begin{aligned} \frac{\partial F(\alpha, \lambda)}{\partial \alpha} &= \left\langle \frac{\partial U(r, \alpha, \lambda)}{\partial \alpha} \right\rangle_\alpha = \left\langle \frac{k\lambda(r - r_0)^4}{2(1 + \alpha(r - r_0)^2)^2} \right\rangle_\alpha \\ &= \lambda \left\langle \frac{k(r - r_0)^4}{2(1 + \alpha(r - r_0)^2)^2} \right\rangle_\alpha = \lambda I(\alpha) \end{aligned}$$

The ensemble average $I(\alpha)$ is a finite number since

$$\begin{aligned} I(\alpha) &= \left\langle \frac{k(r - r_0)^4}{2(1 + \alpha(r - r_0)^2)^2} \right\rangle_\alpha = \int_0^{r_{\text{max}}} dr \frac{k(r - r_0)^4}{2(1 + \alpha(r - r_0)^2)^2} \frac{\exp(-\beta U(r, \alpha, \lambda))}{\int_0^{r_{\text{max}}} dr \exp(-\beta U(r, \alpha, \lambda))} \\ &\leq \int_0^{r_{\text{max}}} dr k(r - r_0)^4 \frac{1}{\exp(-\beta U(r_{\text{max}}, \alpha, \lambda))V} \\ &\leq \frac{kr_{\text{max}}^4}{\exp(-\beta U(r_{\text{max}}, \alpha, \lambda))V} \end{aligned}$$

Therefore,

$$\lim_{\lambda \rightarrow 0} \frac{\partial F(\alpha, \lambda)}{\partial \alpha} = \lim_{\lambda \rightarrow 0} \lambda I(\alpha) = 0$$

Thus,

$$\begin{aligned} F(\alpha = \alpha' > 0, \lambda = 0) - F(\alpha = 0, \lambda = 0) &= \int_0^{\alpha'} d\alpha \lim_{\lambda \rightarrow 0} \frac{\partial F(\alpha, \lambda)}{\partial \alpha} \\ &= \int_0^{\alpha'} d\alpha \lim_{\lambda \rightarrow 0} \lambda I(\alpha) \\ &= 0 \end{aligned}$$

We have proved that the initial states when $\lambda = 0$ for the conventional coupling potential and for the soft bond potential have the same free energy irrespective of the soft bond parameter.

APPENDIX C

Optimal Lambda Schedules for Soft Bond Potential

In a previous work by Shenfeld et. al., the authors have proved that the path with the minimal thermodynamic length is the optimal thermodynamic pathway that minimizes the variance of the calculated free energy, and the optimal lambda schedules are equidistant points on the geodesics.⁴² In this section, we will

prove that the optimal lambda schedules for the soft bond potential are approximately geometric series when λ is not close to 0.

For a thermodynamic path in a one-dimensional space $\lambda(t)$, where $t \in [0, 1]$, and $\lambda(t=0) = 0$ corresponds to the state without the bond and $\lambda(t=1) = 1$ corresponds to the state with the bond, the thermodynamic length along the path is defined as

$$L(\lambda) = \int_0^1 \sqrt{\frac{d\lambda}{dt} g(\lambda) \frac{d\lambda}{dt}} dt$$

where $g(\lambda)$ is the Fisher information matrix at λ

$$g(\lambda) = \left\langle \frac{\partial l_\lambda(x)}{\partial \lambda} \frac{\partial l_\lambda(x)}{\partial \lambda} \right\rangle_\lambda$$

and $l_\lambda(x) = \ln P_\lambda(x)$ is the log of the probability that the system is at configuration x at thermodynamic coupling parameter λ .

For soft bond potential, when the coupling parameter is not close to 0, the distance between the two atoms is close to their equilibrium distance, and the soft bond potential is approximately equal to the harmonic potential with scaled force constant, i.e.

$$U_{\text{sub}}(\lambda, r) = \frac{1}{2} \lambda k(r - r_0)^2 \frac{1}{1 + \alpha(1 - \lambda)(r - r_0)^2}$$

$$\approx \frac{1}{2} \lambda k(r - r_0)^2 \text{ when } |r - r_0| \approx 0$$

Under this approximation, it is easy to show that

$$g(\lambda) = \frac{1}{2\lambda^2}$$

Therefore, the equidistant points along the path are solved by

$$\frac{d\lambda}{dt} g(\lambda) \frac{d\lambda}{dt} = \left(\frac{d\lambda}{dt} \right)^2 \frac{1}{2\lambda^2} = \text{const}$$

The solution of the above equation is

$$\lambda(t) = A \exp(\theta t)$$

We have proved that the equidistant points along the path for scaled harmonic potential are geometric series, and the equidistant points along the path for soft bond potential are approximately geometric series when λ is not close to 0.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge via the Internet at The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jctc.6b00991.

Structures of individual ligands and the protein targets used as the input of the Core Hopping FEP calculations, the output files of all the calculations, the free energy results using the regular R-group modification FEP, and the detailed comparison between Core Hopping FEP and regular R-group FEP results (PDF)

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